# Determinants of Household Participation in Securities Markets

## An Empirical Analysis of the 2025 SEBI Household Survey

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April 2026

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# The Problem

**Only 25.5% of Indian households invest in securities.**

Despite the post-2020 retail boom and zero-commission platforms, participation remains sharply stratified by socioeconomic position, geography, and information access.

**Four research questions:**

1. Who is likely to participate?
2. What keeps the 74.5% out?
3. Which securities do investors prefer, and for how long?
4. Can we predict these outcomes from basic profiles?

**Sample**

| | |
|---|---|
| Total respondents | 109,430 |
| Active investors | 27,882 (25.5%) |
| Non-investors | 81,548 (74.5%) |
| Geographic zones | 4 |
| States / UTs covered | 35 |

**Methods**

Logistic regression (HC3 robust SE, survey-weighted), K-means clustering, Histogram-based Gradient Boosting

# Dataset and Schema

**Source:** 2025 SEBI Household Survey, nationally representative, pre-computed survey weights.

**ETL pipeline:** 449 raw columns transformed into boolean, integer, and ordinal types; loaded into a normalized PostgreSQL schema (17 domain-specific tables anchored by `respondent_id`).

**Conditional routing:**

- Investors: sub-module SS_B3 (info sources, motivations), $n = 9{,}797$
- Non-investors: sub-module SS_BB2 (barriers), $n = 14{,}794$

All group percentages computed against the routing-eligible population.

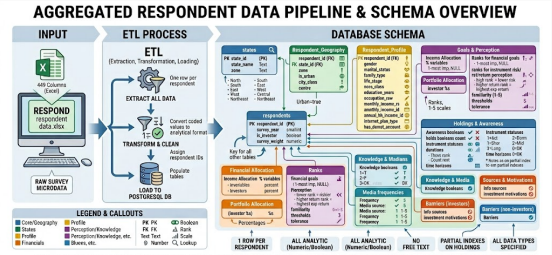


**Figure 1:** *

Schema architecture: modular, indexed, one-to-one with core respondent table.

# EDA: Who Participates?

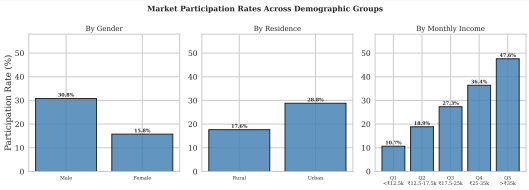


**Figure 2: ***

Participation rates by gender, residence, and monthly income quintile.

**Steepest univariate signals:**

- Income gradient: 10.7% (Q1, <Rs. 12,500/mo) to 47.6% (Q5, >Rs. 35,000/mo): a **4.5**× spread
- Gender gap: 30.8% (male) vs 15.8% (female), persists across all income levels
- Urban premium: 28.8% vs 17.6% (rural)
- Millennials (28–43) lead cohorts at 28.1%, exceeding both Gen Z (24.3%) and Gen X (22.9%)

These univariate signals motivate the multivariate regression: which factors

# EDA: Awareness Gap and Holdings

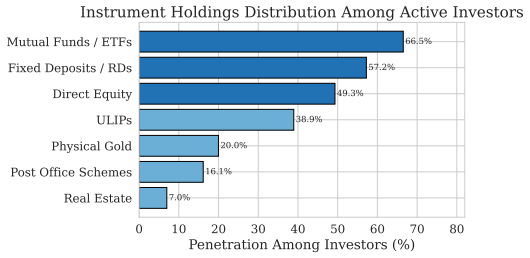


**Figure 3: ***

Instrument holdings among active investors ($n = 27{,}882$).

**Awareness gradient (all 109,430 respondents):**

- Fixed Deposits: **99.1%**
- Mutual Funds / ETFs: 67.0%
- Direct Equity: 62.0%
- Derivatives: **17.9%**

**Holdings among investors:**

- MF/ETF: 66.5%  FD/RD: 57.2%
- Direct equity: 49.3%

MFs and FDs *co-dominate*: Indian retail investors layer market-linked products on top of traditional savings rather than replacing them.

# Finding 1: What Drives Market Entry?

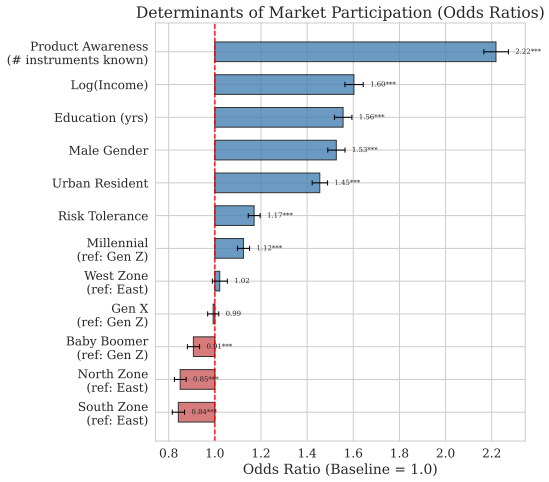


**Figure 4: ***

**Product awareness** is the single strongest predictor (OR = 2.22), exceeding:

| | |
|---|---|
| Income (log) | OR = 1.60 |
| Education | OR = 1.56 |
| Male gender | OR = 1.53 |
| Urban | OR = 1.45 |

Partial mediation: income and education partly work *through* awareness, but awareness retains a large independent effect after SES controls.

Two predictors return null results (Gen X, West zone): the model does not produce spurious significance at $N \approx 100{,}000$.

# Finding 2: Three Archetypes of Non-Participation

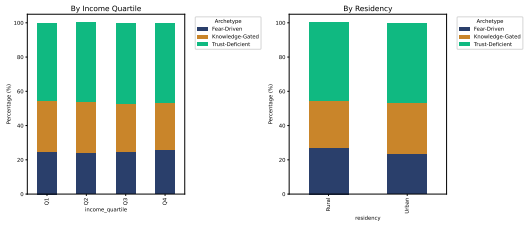


**Figure 5: ***

Demographic distribution of barrier archetypes ($n = 14{,}705$ non-investor barrier responses).

K-means on 18 self-reported barriers identifies three profiles:

**Trust-Deficient (46.3%)**

Persistent across income levels and urban cohorts. Requires institutional accountability: grievance redressal, IPF promotion, broker transparency.

**Knowledge-Gated (29.0%)**

Concentrated in rural areas and lowest income quartile. Needs geographically targeted vernacular literacy delivery.

**Fear-Driven (24.8%)**

Responds to loss-framing education and first-loss protection onramps (SGB, RBI

# Finding 3: Behavioral Traps Inside the Market

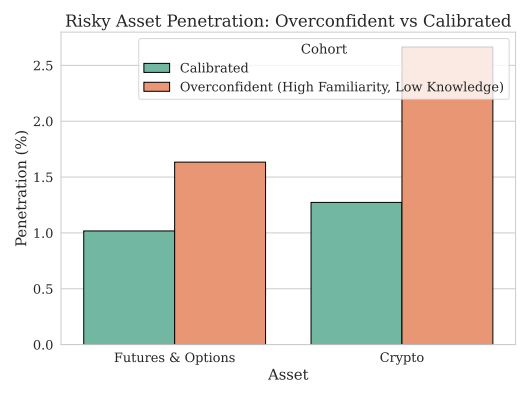


**Figure 6: ***

Overconfident investors hold risky assets at significantly higher rates than calibrated

**Dunning-Kruger effect:** Investors with high perceived familiarity but low objective knowledge hold F&O and crypto at significantly higher rates than calibrated peers.

**Finfluencer pipeline:** 5.0% of investors cite social media as primary information source; this cohort is disproportionately quick-gains oriented relative to professionally-advised investors.

**Goal-portfolio gap:** The "Daily Trading" cohort has a goal-portfolio consistency score of 7.3%: they identify as speculators without holding the instruments that operationalize that strategy.

# Finding 4: Investor Education Programme Impact

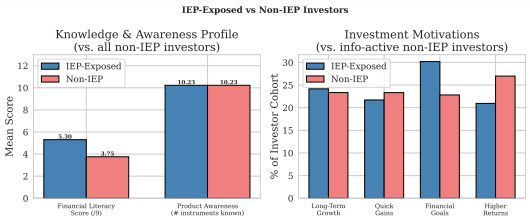


**Figure 7: ***

IEP-exposed vs. non-IEP investors: knowledge, product awareness, investment motives, and holdings.

**2,768 investors (9.9%)** cite SEBI or industry IEP as a top information source.

| Metric | IEP | Non-IEP |
|---|---|---|
| Knowledge score | **5.30** | 3.75 |
| Products aware | 10.23 | 10.23 |
| Goal-based (%) | **30.2** | 22.8 |
| Quick gains (%) | 21.7 | 23.3 |
| Holds FDs (%) | 49.3 | **55.1** |

The 41% knowledge advantage is specific to *conceptual depth*, not product breadth (awareness is near-identical).

IEP functions as both a knowledge-deepening and a philosophical-reorientation intervention.

# Predictive Layer: Three Gradient Boosting Models

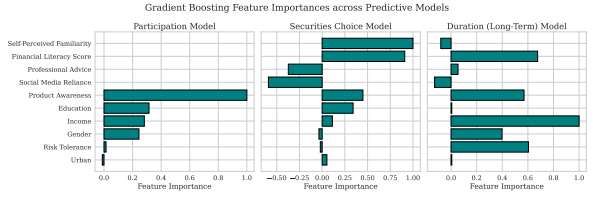


**Figure 8: ***

Normalized permutation feature importances across all three models.

| Model | Task | AUC |
|---|---|---|
| M1 | Participation | **0.814** |
| M2 | Asset choice | **0.820** |
| M3 | Horizon | 0.612 |

**Feature importance shift:**

At the entry layer, *product awareness* dominates. Inside the market, *perceived familiarity* and *objective knowledge* take over.

Investment horizon (M3) is the hardest prediction, consistent with it being driven by situational factors outside a cross-sectional feature space.

# Interactive Dashboard and Planned Chatbot Layer

**Next.js frontend / FastAPI backend**

Six interactive sections:

1. **Geographic Penetration** — investor density map across 35 states with per-capita income overlay
2. **Behavioral Patterns** — Dunning-Kruger and finfluencer charts
3. **Household Profile Simulator** — live three-model prediction from user-adjustable inputs
4. **Participation Frontier** — 10×10 income × education probability heatmap
5. **Asset Ladder** — holdings penetration by income quartile
6. **Barrier Wall** — proportional mosaic of

**Planned: Chatbot Tab**

An additional tab to be added to the existing dashboard that will allow natural language queries over the survey findings, model outputs, and policy recommendations.

**Planned architecture:**

- LLM with tool-use over the FastAPI endpoints
- Context grounded in the report findings and CSV outputs
- Queries like: "What is the participation probability for a rural woman with 10 years of education?" resolved via live model calls

# Policy Recommendations

**A. Prioritize product awareness**
OR = 2.22; campaigns should name specific instruments (MF, ETF, SIP) for North and South zones.

**B. Segment non-investor interventions**
Trust-Deficient: regulatory transparency.
Knowledge-Gated: vernacular rural delivery.
Fear-Driven: first-loss protection onramps.

**C. Scale IEP reach**
From 9.9% to employer mandates and Jan Dhan partnerships. Content should emphasize conceptual depth, not procedural literacy (KYC awareness is already high). Goal-aligned investors show 72–77% portfolio consistency, confirming goal-based framing works.

**D. Close the gender gap**
OR = 1.53 for male gender after full controls. SHG-linked investment collectives, women-targeted IEP content, lower minimum investment thresholds.

**E. Regulate finfluencers**
5% usage associated with quick-gains orientation. Mandatory qualification disclosure, prohibited performance claims, platform liability for unregistered advice.

**F. Gate derivatives on objective literacy**
Overconfident investors and aspirational daily traders (7.3% consistency) reach leveraged instruments without adequate knowledge. Replace self-certification with a brief objective literacy assessment.

# Appendix: Holdings by Income Quartile

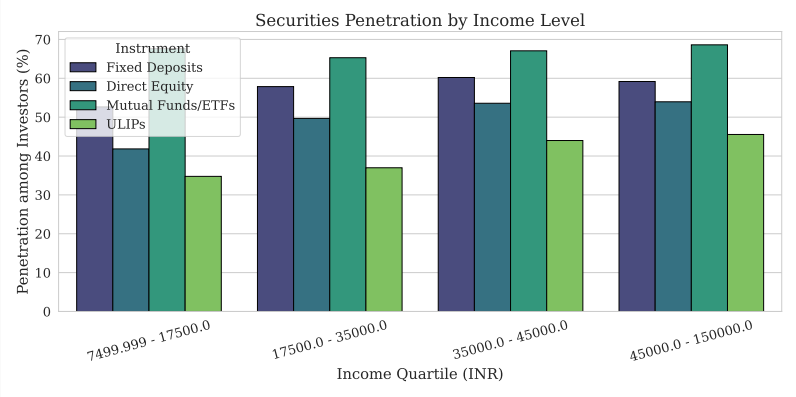


**Figure 9: ***

Market-linked instruments scale sharply with wealth.

# Appendix: Goal-Portfolio Consistency

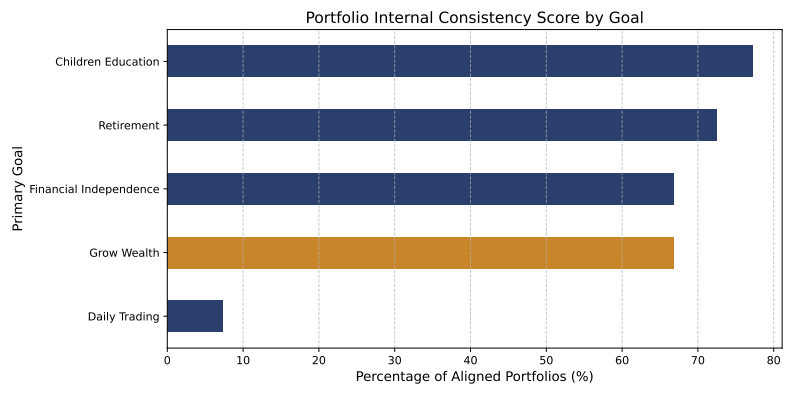


**Figure 10: ***

Consistency score by stated primary goal. Daily Trading cohort: 7.3%.

# Appendix: Investment Horizons by Asset Class

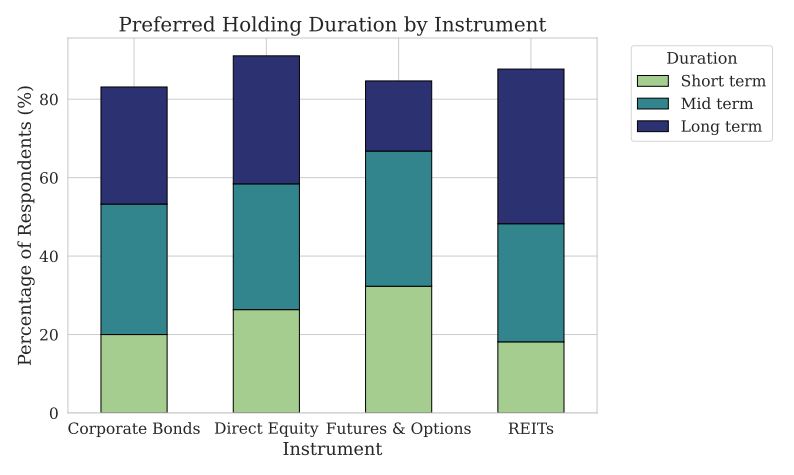


**Figure 11: ***

# Appendix: Full Logistic Regression Table

| **Predictor** | **OR** | **95% CI** | $z$ | Sig. |
|---|---|---|---|---|
| Product Awareness | 2.218 | [2.166, 2.272] | 65.2 | *** |
| Log(Income) | 1.603 | [1.564, 1.643] | 37.4 | *** |
| Education (yrs) | 1.556 | [1.518, 1.594] | 35.6 | *** |
| Male | 1.526 | [1.490, 1.564] | 33.9 | *** |
| Urban | 1.455 | [1.422, 1.489] | 32.0 | *** |
| Risk Tolerance | 1.170 | [1.145, 1.196] | 14.0 | *** |
| Millennial | 1.124 | [1.098, 1.151] | 9.8 | *** |
| Gen X | 0.993 | [0.969, 1.017] | −0.6 | |
| Baby Boomer | 0.907 | [0.881, 0.934] | −6.6 | *** |
| North | 0.850 | [0.825, 0.876] | −10.7 | *** |
| South | 0.842 | [0.816, 0.869] | −10.7 | *** |
| West | 1.021 | [0.989, 1.054] | 1.3 | |

$N = 97{,}626$ McFadden $R^2 = 0.290$ *** $p < 0.001$, HC3 robust SE